

DATA SET

DS1406

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married		
	Yes	No	
Bachelors or higher	187	113	300
Some college	162	138	300
High school or less	149	151	300
	498	402	900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

DATA SET

DS1407

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence			
	Apartment	Duplex	Single Residence	
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or More	6	2	16	24
	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

10.

H_0 : Marriage rates 和 education levels 獨立

H_a : Marriage rates 和 education levels 相關

$$E_{Yes} = \frac{300 \times 498}{900} = 166$$

$$E_{No} = \frac{300 \times 402}{900} = 134$$

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 10.06 > \chi^2_{0.01, 2} = 9.21$$

$\therefore$ Reject H_0 .

在 $\alpha=0.01$ 下 Marriage rate 和 education levels 顯著相關, education levels 越高, marriage rate 越高

12.

H_0 : family size 和 types of residence 獨立

H_a : family size 和 types of residence 相關

$$E_1 = \frac{29}{3} = 9.667, E_2 = \frac{33}{3} = 11, E_3 = \frac{34}{3} = 11.333, E_4 = \frac{24}{3} = 8$$

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 36.999 > \chi^2_{0.01, 6} = 16.812$$

$\therefore$ Reject H_0

在 $\alpha=0.01$ 下, different types of residence 的家庭人數分布存在顯著差異。

① Duplex 主要以 1 人家庭為主 (50%) ② Single Residence 集中在 3 人、4 人 ③ Apartment

相對平均

C415

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

上游	2	10	8	6	9
Above town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

- Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.
- Use a Student's t-test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

12. 若 Town 排放污水, 下游溶氧量
應顯著低於上游 $\rightarrow$ One-tailed test.

(a) Wilcoxon Rank Sum Test.

H_0 : 上游、下游溶氧量完全相同

H_a : 上游溶氧量 $>$ 下游溶氧量

Rank: 4.8 平均排名: $\frac{2+3}{2} = 2.5$
 4.9 " $\frac{4+5+6}{3} = 5$
 5 " $\frac{7+8}{2} = 7.5$

② 上游 $T_1 = 2.5 + 10 + 7.5 + 5 + 9 = 34$

下游 $T_2 = 7.5 + 1 + 5 + 2.5 + 5 = 21$

③ $n_1 = n_2 = 5$, $T_0 = 19$, R.R.: reject $T < 19$

$\therefore T_2 = 21 > 19 \therefore$ non-reject H_0

(b) t-test.

$$\textcircled{1} \bar{x}_1 = 5 \quad S_1^2 = 0.045$$

$$\bar{x}_2 = 4.86 \quad S_2^2 = 0.013$$

$$\textcircled{2} S_p^2 = \frac{(5-1)S_1^2 + (5-1)S_2^2}{5+5-2} = \frac{0.1 + 0.052}{8} = 0.019$$

$$S_p = 0.13784$$

$$\textcircled{3} t = \frac{\bar{x}_1 - \bar{x}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{5 - 4.86}{0.13784 \sqrt{\frac{2}{5}}} = 1.606$$

$$t_{0.05, 8} = 1.86$$

$\therefore 1.606 < 1.86 \therefore$ non-reject H_0

$$P(T \leq 19) \leq 0.05$$

意思是說：「如果上游和下游的水質真的完全一樣 (H_0 為真)，那麼因為隨機抽樣的誤差，導致下游的排名加總 (秩和) 小於或等於 19 的機率，只有不到 5%

($\alpha = 0.05$)。」「統計學課本.pdf +1

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

Deaf Children	Hearing Children
2.75 (15)	.89 (1)
2.14 (11)	1.43 (7)
3.23 (18)	1.06 (4)
2.07 (10)	1.01 (3)
2.49 (14)	.94 (2)
2.18 (12)	1.79 (8)
3.16 (17)	1.12 (5.5)
2.93 (16)	2.01 (9)
2.20 (13)	1.12 (5.5)
Rank Sum	126

Two Simple Examples Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

13. H_0 : 兩組兒童的 eye-movement rate 分布完全相同.

H_a : 兩組兒童的 eye-movement rate 分布 different.

$$T_1 = \min(T_{\text{deaf}}, T_{\text{hearing}}) = 45$$

$$T_1^* = h_1(n_1 + n_2 + 1) - T_1 = 9 \times (9 + 9 + 1) - 45 = 126$$

$$\text{統計量 } T = \min(T_1, T_1^*) = 45$$

$$T_0(9, 9, 0.05) = 62$$

$\because T = 45 < 62 \therefore \text{Reject } H_0.$

Deaf children 和 Hearing children 的 eye-movement rate 分布存在顯著差異